Task 1 – Development of the Blade Structural Optimisation Code

1 Starting point: a solver wrapper

The brief supplies `beamBending`, an Euler–Bernoulli cantilever solver that takes the second moment of area I at $N = 100$ spanwise nodes and returns the deflection $\delta(r)$ and bending moment $M(r)$ under the worst-case load $L_z(r) = 50r\,(1 - e^{50(r-R)})$. Everything else (the cross-section, the volume and the stress check) had to be built around it. The first script, `RunBeamBending.m`, therefore has one job: turn a structural concept into the vector $I(r)$ and the area $A(r)$, call the solver once, and report the volume $V = \int_0^R A\,\mathrm{d}r$, the tip deflection and the peak bending stress $\sigma = |M|\,y_{\max}/I$. Following the "start simple" advice, the first concept was a single solid spar whose width tapers linearly from root to tip.

2 Choosing the structural concept: spar vs. skin

Two section models were coded behind a single switch, so that both concepts go through the same solver, stress check and plots:

- **Approach 1 – solid spar.** The spar, centred at $x/c = 0.3$ (maximum thickness), is split into 50 vertical strips of height $2y_t(x)$ from the NACA 0012 equation, giving $I = \sum b_i d_i^3/12$ and $A = \sum b_i d_i$.
- **Approach 2 – thin structural skin** ($t \ll c$). Each surface is split into 200 straight panels with cosine spacing, which concentrates panels round the strongly curved leading edge. Each panel contributes $L^3 t \sin^2\theta/12 + (Lt)\,d^2$ (parallel axis theorem); the lower surface mirrors the upper.

A key observation from coding Approach 2 is that the skin's section properties are *linear in thickness*: $I = I_u\,t$ and $A = P\,t$, where I_u and the perimeter P are fixed by the aerofoil shape. They are computed once and then scaled.

With hand-picked linear tapers sized to sit just under yield, the skin carried the load with $367\,\mathrm{cm}^3$ of material against $535\,\mathrm{cm}^3$ for the spar (31 % less), at the same peak stress of 29.6 MPa. This is what the definition $I = \int y^2\,\mathrm{d}A$ predicts: the skin places all of its material at the outer fibre, whereas the solid spar puts much of its material near the neutral axis, where it adds area but little stiffness. The skin was therefore taken forward to optimisation.

3 Making each evaluation almost free

The brief asks for a tool that can be rerun for thousands of regions, so evaluation cost mattered more than anything else. Two properties of the problem remove the solver from the optimisation loop entirely:

1. **The blade is statically determinate.** $M(r)$ follows from equilibrium alone and does not depend on I. One call to `beamBending` (with any I) gives $M(r)$ for *every* design.
2. **Unit-load (virtual work) method.** With M known, the tip deflection is a single integral,

$$\delta_{\mathrm{tip}} = \int_0^R \frac{|M(r)|\,(R-r)}{E\,I(r)}\,\mathrm{d}r, \tag{1}$$

   where $(R - r)$ is the moment due to a unit tip load. It agrees with the full solver to within 0.02 % (e.g. 0.6282 m vs. 0.6281 m); this check is rerun automatically at three points on every front.

Together these turn each design evaluation into a handful of vector operations. The complete Pareto front of 15 optimisations takes 1.3 s and needs exactly one solver call.

4 Formulating the optimisation

The skin thickness was parameterised by its root and tip values, $t(r) = t_{\mathrm{root}} + (t_{\mathrm{tip}} - t_{\mathrm{root}})\,r/R$: two variables, consistent with the bending moment falling from root to tip. The brief asks for minimum material, while a turbine blade also has to stay stiff. Rather than combine the two with arbitrary weights, the trade-off was mapped as a Pareto front using the ε-constraint method, which gives evenly spaced points and, unlike a weighted sum, would also capture a non-convex front:

$$\min_{t_{\mathrm{root}},\,t_{\mathrm{tip}}} V \quad \text{s.t.} \quad \frac{\sigma_{\max}}{\sigma_{\mathrm{allow}}} - 1 \le 0, \qquad \frac{\delta_{\mathrm{tip}}}{\varepsilon} - 1 \le 0, \qquad 0.8\,\mathrm{mm} \le t \le 2.0\,\mathrm{mm}. \tag{2}$$

Each decision has a stated reason:

- $\sigma_{\mathrm{allow}} = 30\,\mathrm{MPa}/1.5 = 20\,\mathrm{MPa}$. The safety factor covers print-quality scatter and the unmodelled shear stresses.

- The upper bound keeps $t \lesssim 20\,\%$ of the 9 mm half-depth, so the thin-wall model stays valid.
- The lower bound is two 0.4 mm printed perimeters, the minimum for a structural wall.
- Constraints are normalised, so that `fmincon` sees terms of order one.

Algorithm 1 Pareto front generation (`ParetoSkin.m`)

$M(r) \leftarrow$ `beamBending`$(N,\, I_{\text{dummy}})$ *only solver call*
$\mathbf{x}_{\text{light}} \leftarrow \arg\min V$ s.t. stress only; $\delta_{\text{light}} \leftarrow \delta(\mathbf{x}_{\text{light}})$ *anchor 1*
$\delta_{\text{stiff}} \leftarrow \delta(t_{\max}, t_{\max})$; $\mathbf{x}_0 \leftarrow (t_{\max}, t_{\max})$ *anchor 2*
for ε in `linspace`$(\delta_{\text{stiff}}, \delta_{\text{light}}, 15)$ **do**
    $\mathbf{x}^* \leftarrow$ `fmincon`$(V, \mathbf{x}_0,$ stress & $\delta \le \varepsilon)$; $\mathbf{x}_0 \leftarrow \mathbf{x}^*$ *warm start*
    $\mathbf{x}_{\text{print}} \leftarrow$ round $t(r)$ *up* to multiples of 0.4 mm
end for
Validate: compare δ with `beamBending`; plot the front over a 30×30 brute-force grid

5 Designing for the printer

A continuous optimum such as $t = 1.78$ mm cannot be printed: an FDM wall is a whole number of 0.4 mm extrusion lines. Each optimum is therefore rounded *up* to the next perimeter. Rounding up only ever adds material, so the stress constraint stays satisfied and the deflection can only fall, and no re-optimisation is needed. Two printable designs are compared with the bounds in Table 1. The continuous front is the real physical trade-off; the printable front (blue in Fig. 1) is what the company can actually make.

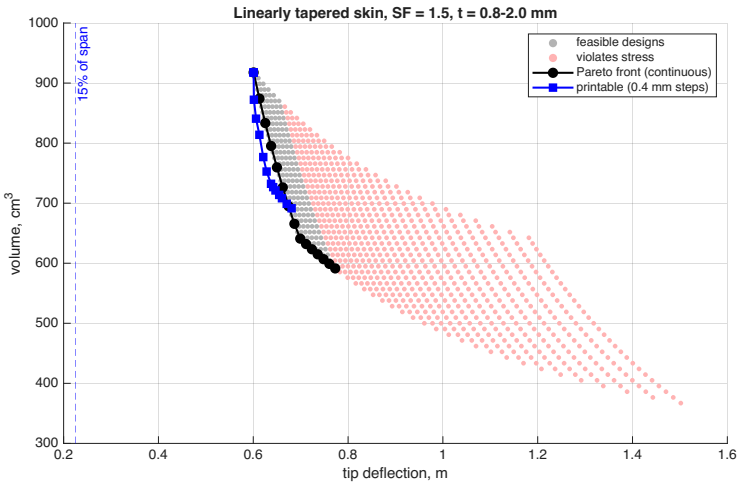


Figure 1: Volume vs. tip deflection for the linearly tapered skin. Grey and red dots: brute-force grid (stress-feasible and infeasible). Black: continuous ε-constraint front, which lies on the edge of the feasible grid and so confirms that `fmincon` converged to the true optima. Blue: printable designs after rounding up to whole perimeters.

Table 1: Bounds of the continuous front (top) and two printable designs (bottom).

Design	V (cm^3)	δ_{tip} (m)	$t_{\text{root}}/t_{\text{tip}}$ (mm)
Lightest	591	0.77	1.78 / 0.80
Stiffest	918	0.60	2.00 / 2.00
Lightest	692	0.68	2.0 / 0.8
Point 8	753	0.63	2.0 / 1.2

6 What the results show so far

- **The optima behave as expected.** Along the front, material is first removed from the tip, where the moment is small. Once t_{tip} reaches the 0.8 mm print limit ($V \approx 641$ cm^3), only the root can thin further, which creates the kink in the front. At the lightest point the stress constraint is exactly active at the root.
- **Printing costs material.** Rounding to whole perimeters adds 17 % to the lightest design (591 → 692 cm^3). It also collapses the three stiffest points onto one design, because 1.6–2.0 mm all round to the same perimeter count. This rounding loss is now one of the largest sources of inefficiency.
- **Stiffness is the limiting physics.** Every design deflects 0.60–0.77 m, 40–50 % of the span. This is far from the 15 % guideline and outside the small-deflection validity of Euler–Bernoulli theory. A thin ABS skin limited to 2 mm is strength-adequate but not stiff enough.

7 Next steps

- **Non-linear thickness distributions.** Try a power law and pchip control points. Because $I \propto t$, the problem is convex, and the optimality conditions give $t^*(r) \propto \sqrt{|M|(R-r)}$, clipped by the stress and print limits. This is an analytic benchmark that any parameterisation can be measured against.
- **Optimise the number of perimeters per zone directly.** Treating perimeter counts as integers (`ga` with `IntCon`) would remove the rounding loss.
- **Address the stiffness deficit.** Relax $t_{\max}$ with a thick-wall section, or combine the skin with a spar (Approach 1 + 2).